Econometrics

Sampling Distribution and Central Limit Theorem

Random Sampling

- Imagine casting a die twice and deriving the mean. Let X_1 be the number from 1st trial and X_2 from 2nd, each trial has the exactly same distribution as the population.
- Sometimes we may observe (3, 5) and the sample mean will be 4 in this case.
- Before observing these values, we don't know what the sample mean will be. The point is that the sample mean $\frac{X_1+X_2}{2}$ will be another **random variable**.
- Now, QUESTION: **what will be the distribution of the sample mean?**

Random Sampling

- Since $\{X_1, X_2\}$ are selected at random from a population, the sample mean $\overline{X} = \frac{X_1+X_2}{2}$ is a statistic, but random statistic (random variable).
- Since $\overline{X} = \frac{X_1+X_2}{2}$ is a random variable, there exist its own distribution.
- The distribution is called "Sampling Distribution".
- Characterizing the distributions of sample averages is an essential step toward understanding the performance of econometric procedures.

Random Sampling

- Because $X_1, \cdots, X_n$ (n observations) are randomly drawn from the same population, the marginal distribution of X_i is the same for each observation; this marginal distribution is the distribution of X in the population being sampled.
- Under simple random sampling, knowing the value of X_i provides no information about X_j.
- When $X_1, \cdots, X_n$ are drawn from the same distribution and are independently distributed, they are said to be **independently and identically distributed (i.i.d.)**

The Sampling of the Sample Average

- Suppose that the observations $X_1, \cdots, X_n$ are *i.i.d.*, and let μ_X and σ_X^2 denote the mean and variance of X_i. Thus, the sample average or sample mean is:

$$E(\bar{X}) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \mu_X$$

- The variance of $\bar{X}$ is:

$$Var(\bar{X}) = Var(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{\sigma_X^2}{n}$$

- The standard deviation of $\bar{X}$ is $\sigma_X^2/\sqrt{n}$.

The Sampling of the Sample Average

- Suppose that $X_1, \cdots, X_n$ are *i.i.d.* draws from the $N(\mu, \sigma_X^2)$ distribution.
- The sum of n normally distributed random variables is itself normally distributed.
- Because the mean of $\bar{X}$ is μ and the variance of $\bar{X}$ is σ_X^2/n, $\bar{X}$ is distributed $N(\mu, \sigma_X^2/n)$.

Large-Sample Approximations to Sampling Distributions

- There are two approaches to characterizing sampling distributions: an "exact" approach and an "approximate" approach.
- The exact approach entails deriving a formula for the sampling distribution that holds exactly for any value of n.
  - For example, if X is normally distributed and $X_1, \cdots, X_n$ are *i.i.d.*, then the exact distribution of $\bar{X}$ is also normal.
- The sampling distribution that exactly describes the distribution of $\bar{X}$ for any n is called the **exact distribution** or **finite-sample distribution**.

Large-Sample Approximations to Sampling Distributions

- Unfortunately, if the distribution of X is not normal, then in general the exact sampling distribution of $\bar{X}$ is complicated and depends on the distribution of X.
- Instead, the approximate approach uses approximations to the sampling distribution that rely on the sample size being large.
- The large-sample approximation to the sampling distribution is often called the **asymptotic distribution**—"asymptotic" because the approximations become exact in the limit that $n \to \infty$.

Large-Sample Approximations to Sampling Distributions

- Below we will present the two key tools used to approximate sampling distributions when the sample size is large: the law of large numbers and the central limit theorem.
- We could see that although exact sampling distributions are complicated and depend on the distribution of X, the asymptotic distributions are simple based on the two tools.

The Law of Large Numbers

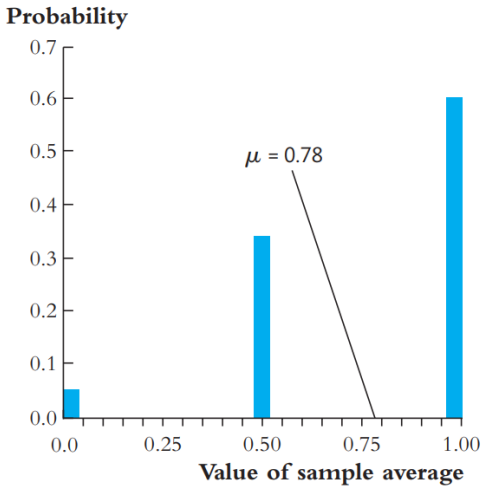
- The **law of large numbers** states that, under general conditions, $\bar{X}$ will be near μ_X with very high probability when n is large.
- The property that $\bar{X}$ is near μ_X with probability increasing to 1 as n increases is called **convergence in probability** or, more concisely, **consistency**:

$$\bar{X} \xrightarrow{p} \mu_X$$

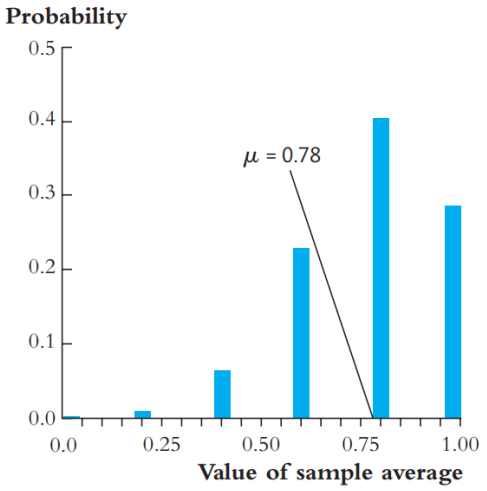
The Law of Large Numbers

- The property that $\bar{X}$ is near μ_X with probability increasing to 1 as n increases is called **convergence in probability** or, more concisely, **consistency**.
- Note that the conditions for the law are that $X_1, \cdots, X_n$ are $i.i.d$ and that the variance of X_i, is finite or large outliers are unlikely (technically if $Var(X_i) = \sigma_X^2 < \infty$).

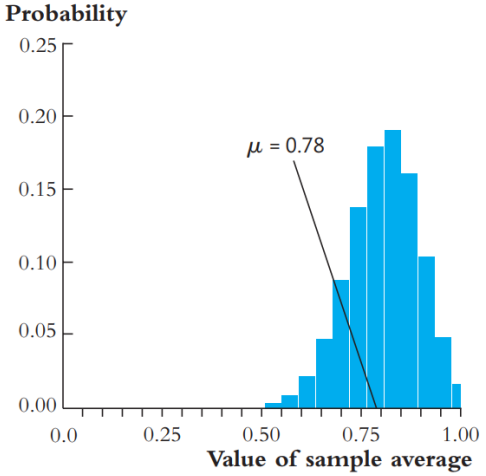
Exp. The Law of Large Numbers



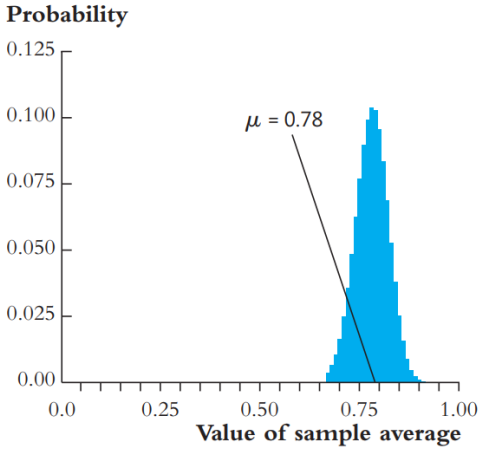
(a) $n = 2$



(b) $n = 5$



(c) $n = 25$



(d) $n = 100$

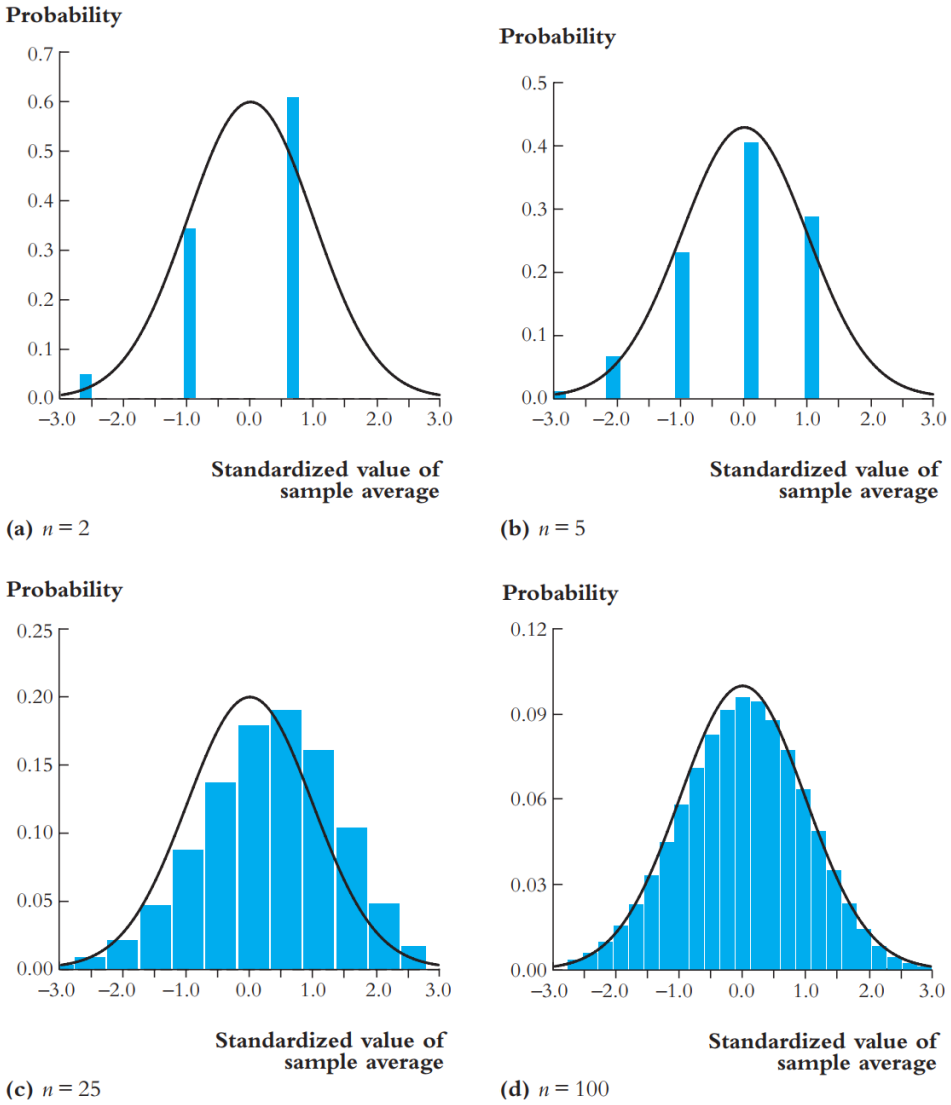
The Central Limit Theorem

- The **central limit theorem** says that, under general conditions, the distribution of $\bar{X}$ is well approximated by a normal distribution when *n* is large.
- According to the CLT, when *n* is large, the distribution of $\bar{X}$ is approximately $N(\mu_X, \sigma_{\bar{X}}^2)$, the same with the case of exact distribution of $\bar{X}$.
- In other words, the CLT says that this same result is *approximately* true when *n* is large even if $X_1, \cdots, X_n$ **are not** themselves normally distributed.

The Central Limit Theorem

- It would be easier to see the shape of the distribution of $\bar{X}$ to standardize $\bar{X}$ (i.e., $(\bar{X} - \mu_X)/\sigma_{\bar{X}}^2$), so that is has a mean of 0 and a variance of 1.
- According to the CLT, this distribution should be well approximated by a $N(0, 1)$ distribution when n is large.
- Overall, in either case, $\bar{X}$ is said to have an **asymptotic normal distribution**.

Exp. The Central Limit Theorem



The Central Limit Theorem

- One might ask, how large is "large enough"? That is, how large must n be for the distribution of $\bar{X}$ to be approximately normal?
- The answer is, "It depends." The quality of the normal approximation depends on the distribution of the underlying X_i that make up the average.
- In many practical situations, a sample size of 30 may be sufficiently large.

The Central Limit Theorem

- The central limit theorem is summarized in follow:

  Suppose X_i is a sequence of iid random variables with $E(X_i) = \mu_X$, and $Var(X_i) = \sigma_X^2 < \infty$, then as $n \to \infty$, we have

$$\sqrt{n}(\bar{X} - \mu_X) \to_d N\left(0, \sigma_X^2\right)$$